

Jackson Bean

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EDUCATION

The Ohio State University

B.S. in Computer Science and Engineering and Theoretical Mathematics

August 2021 – May 2026

Columbus, OH

GPA: 4.00/4.00

EXPERIENCE

Student Instructional Associate

The Ohio State University

August 2024 – May 2026

Columbus, OH

- Prepared and taught College Algebra recitation twice a week to over 70 students divided into two sections.
- Answered over 20 questions each week about course material in a concise and understandable manner.
- Graded and proctored 10 weekly quizzes and four monthly exams over semester-long course content.

Cognitive Systems Engineering Research Intern

The Ohio State University

May 2024 – September 2024

Columbus, OH

- Worked with the Cognitive Systems Engineering Lab (CSEL) to develop two new contingency plans for UAMs.
- Developed C++ code to run more than 40 simulations that enacted parameter-determined flight pathing.
- Analyzed results of simulations using pandas and Pyplot Dash in Python to create six dynamic charts and graphs.

Biomedical Informatics Research Intern

The Ohio State University

May 2023 – August 2023

Columbus, OH

- Constructed and trained an AI model using gradient boosting, random forests, and SVMs with scikit-learn in Python that classified patient outcomes with more than 80% accuracy.
- Utilized pandas and NumPy to efficiently access over one million instances of patient data for training.
- Compared over 150 instances of training and test results using matplotlib in Python and presented findings at a departmental research forum.

PROJECTS

Symplectic Integrators Simulation (OSU Cycle) | *Python*

September 2025 – April 2026

- Analyzed symplectic, area-preserving numerical methods used to simulate interactions between two Argon atoms.
- Studied modified and computational differential equations to represent complex real-world Hamiltonian systems.
- Developed a Python simulation which measured the energy conservation of general numerical methods against symplectic methods.

Personal Portfolio Website | *React, TypeScript, Tailwind CSS*

June 2025

- Developed a dynamic personal portfolio website using industry-standard tools for front-end web development.
- Implemented reusable, stateful React components to organize content for a cohesive user experience.
- Designed and styled application components directly using Tailwind CSS.

Full-Stack Employee Database Web Application | *React, TypeScript, Express.js, MongoDB*

March 2025

- Built a RESTful API backend to access a cloud-based database through MongoDB.
- Applied CRUD principles to a TypeScript-based backend through custom GET, POST, and DELETE routes.
- Programmed and styled a front-end webpage using React and Tailwind CSS for user database searches.

SKILLS

Languages: Python, JavaScript, TypeScript, C/C++, Java, SQL, HTML/CSS, Ruby, x86 Assembly, R, LaTeX

Frameworks & Libraries: NumPy, pandas, scikit-learn, Flask, Express.js, Ruby on Rails, React, Tailwind CSS

Developer Tools: MongoDB, SQLite, Linux/UNIX, Makefile, Git, Vite, VS Code, Xcode, Postman, Docker

Mathematics Experience: Probability, Graph Theory, Parameter Estimation, Real Analysis, Topology, Abstract Algebra, Numerical Analysis, ODEs, PDEs, Linear Algebra, Calculus I-III, Knot Theory

Miscellaneous: Personal Computer Building, Global Seal of Biliteracy in Spanish